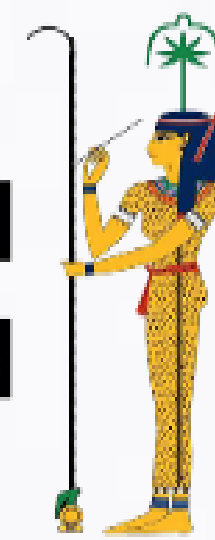




وزارة الاتصالات
وتكنولوجيا المعلومات



Presented by:
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ANUBIS X

A Threat-First Intelligent Surveillance System

Brigadier General Dr.
Tarek Ghoniemy

ABSTRACT

ANUBIS X is an AI-powered weapon threat detection system that analyzes CCTV footage in real time to identify weapon-related threats and track suspicious individuals across multiple cameras.

It combines deep learning for detection and re-identification with a structured workflow to deliver accurate, explainable, and actionable security intelligence via a mobile app.

Our goal is to enhance situational awareness, support rapid response, and improve decision-making for security operations in high-risk environments.



AIM OF THE WORK



Detect actionable weapon threats.



Associate each threat with the most plausible person.



Track only threat-linked individuals.

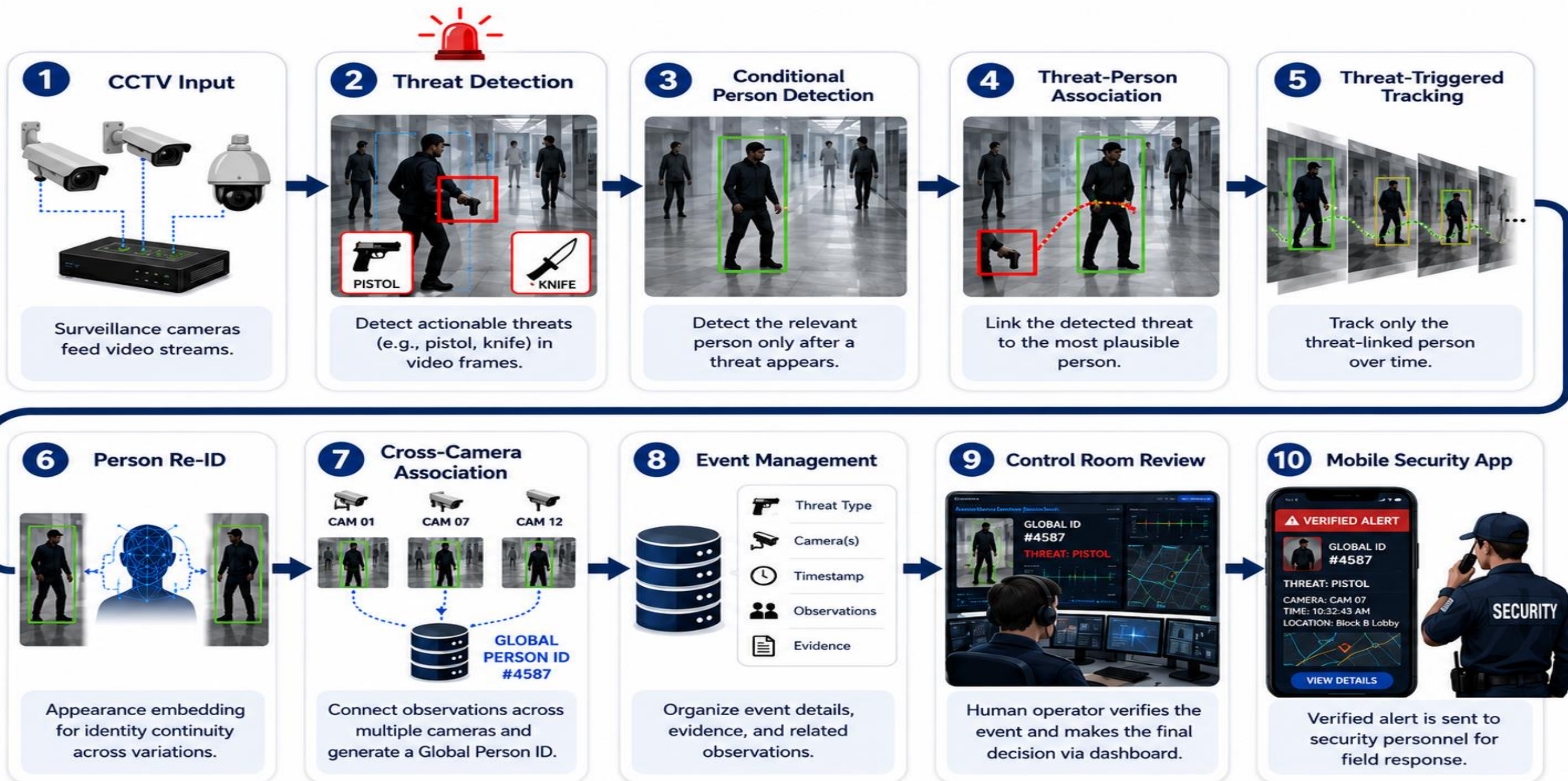


Maintain identity continuity using tracking and Re-ID.



Deliver verified alerts to security personnel via mobile app.

SYSTEM OVERVIEW



RESULTS

THREAT DETECTION

86.3%
Precision

89.8%
Recall

94.1%
mAP@0.5

79.2%
mAP@0.5:0.95

Confusion Matrix (Normalized)

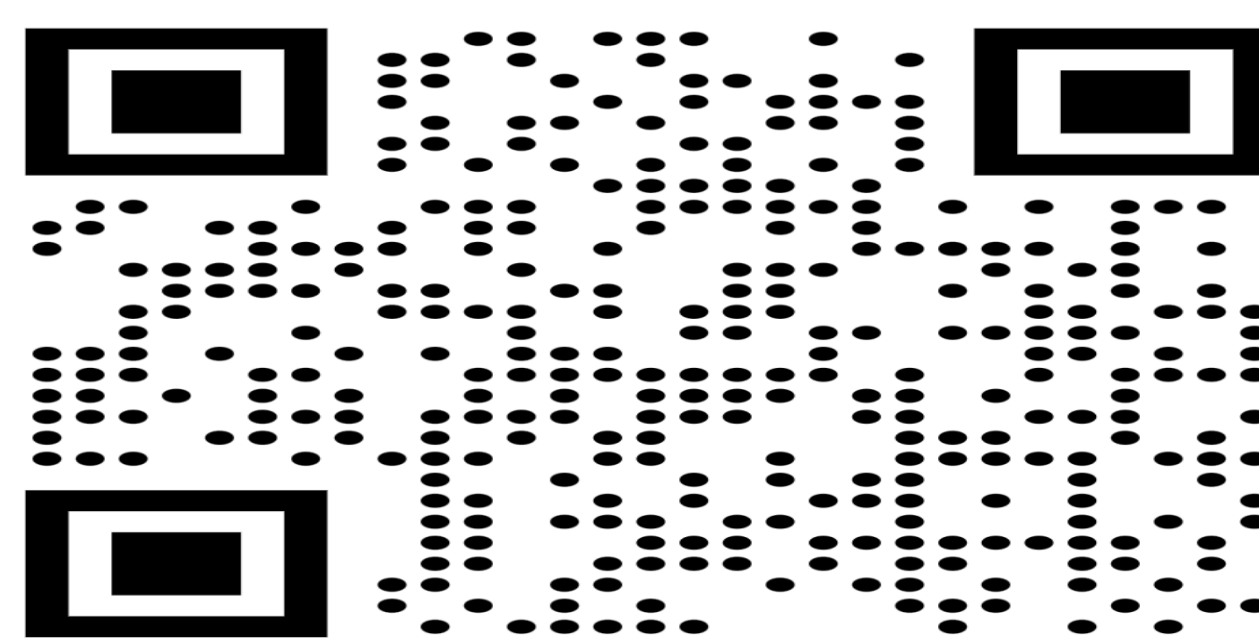
Predicted \ True	True						
	pistol	smartphone	knife	machete	bottle	toygun	background
pistol	0.89	0.03	0.02	0.03	0.00	0.07	0.24
smartphone	0.03	0.87	0.01	0.02	0.01	0.14	0.18
knife	0.03	0.03	0.96	0.05	0.00	0.12	0.22
machete	0.02	0.02	0.02	0.83	0.03	0.12	0.13
bottle	0.06	0.01	0.04	0.03	0.87	0.15	0.06
toygun	0.05	0.01	0.00	0.01	0.11	0.82	0.06
background	0.02	0.02	0.02	0.03	0.03	0.02	0.00

OSNet Re-ID



Rank-1: 48.3%
Rank-5: 71.5%
mAP: 59.5%

DEMOS & PRESENTATION



System Demo
Presentation
Project Evidence

DATASET

SOHAS Weapon-Detection Dataset



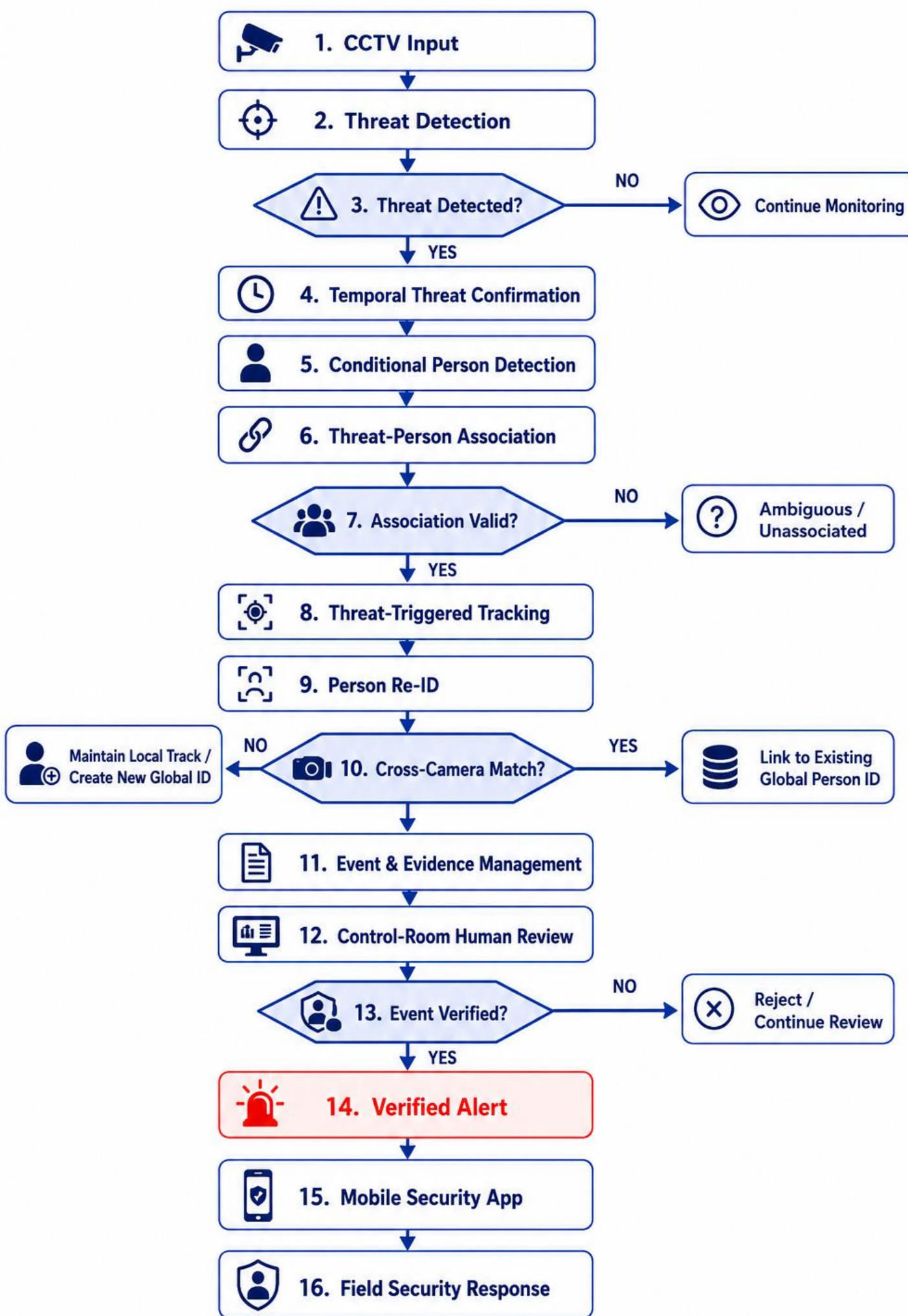
6 Classes



USED TOOLS



METHODOLOGY



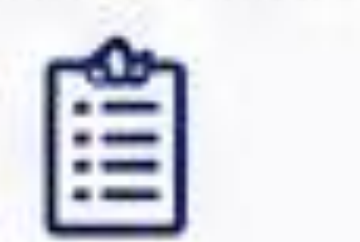
CONCLUSION



From threat detection to verified response, ANUBIS X transforms CCTV footage into structured and actionable security evidence while keeping final decisions under human control.



Threat-First Workflow



Structured Evidence



Human Verification



Actionable Response

FUTURE WORK

